THE TIMES OF INDIA | ENVIRONMENT

TOI+ EXCLUSIVE

HIMALAYAN ECOLOGY & CLIMATE RISK

‘The tourism boom in the Himalayas needs a rethink’

Yadu Pokhrel, an expert on hydroclimatic hazards and professor at Michigan State University, warns that haphazard development and increasing tourism pressure could compound the vulnerability of the Himalayan region to extreme events.

Author: Jayanta Kalita **Publisher:** TIMESOFINDIA.COM **Published:** Sep 2, 2026, 19:29 IST

Topic: Himalayan Eco-Tourism & Hydroclimate Hazards

Haphazard commercial development, unmanaged mass tourism, and rapid construction along Himalayan river corridors amplify devastating flash floods and ice-rock collapses.

KEY ENVIRONMENTAL WARNINGS & RECOMMENDATIONS

- **Cascading Hydroclimatic Hazards:** Recent devastating flash floods (such as the August 2026 Nepal events) demonstrate how high-mountain ice-rock collapses quickly turn into destructive mudslides downstream.
- **Compounding Pressure of Tourism:** Unchecked construction of hotels, roads, and commercial infrastructure in high-altitude risk zones severely destabilizes riverbanks and fragile mountain slopes.

- **Lack of Transboundary Preparedness:** Rivers originating in Tibet/Nepal cross borders into India, requiring synchronized, cross-border early warning systems and real-time hydrological data sharing.
- **Urgent Need for Carrying Capacity Studies:** Himalayan tourist hubs must establish scientifically backed visitor limits to prevent ecological collapse.

The devastating flash floods that struck Nepal in August 2026 have once again exposed the growing hazards posed by extreme and cascading weather events across the fragile Himalayan ecosystem.

Speaking in an exclusive interview with *The Times of India*, **Prof. Yadu Pokhrel**, an expert on hydroclimatic hazards and professor of civil and environmental engineering at Michigan State University, issued a stern warning: haphazard infrastructure expansion and unbridled mass tourism are accelerating mountain ecosystem collapse.

Understanding "Cascading Disasters" in the High Himalayas

Unlike conventional riverine flooding caused purely by heavy rainfall, disasters in the Hindu Kush Himalaya region frequently originate high up in glacierized zones as compound, multi-hazard events.

> "A disaster that begins as a high-altitude ice or rock collapse high up in the mountains can rapidly transform into a massive slurry of mud, boulders, and water. By the time it reaches river valleys, it wipes out entire towns and hydroelectric dams downstream."
>
> — PROF. YADU POKHREL, MICHIGAN STATE UNIVERSITY

According to Pokhrel, when rapid glacier retreat triggered by global warming combines with intense cloudbursts, fragile glacial lakes overflow (Glacial Lake Outburst Floods - GLOFs) or trigger landslide dams that breach catastrophically.

The Uncalculated Cost of Himalayan Mass Tourism

In recent years, regions across Himachal Pradesh, Uttarakhand, Sikkim, and Nepal have witnessed unprecedented tourism surges. However, urban planning and waste management have failed to keep pace.

Environmental Stressor	Impact on Himalayan Ecosystem	Recommended Policy Rethink
River Corridor Encroachment	Hotels, guest houses, and parking lots built directly on active floodplains restrict natural water channels.	Enforce strict 100-meter buffer zones and ban permanent structures on riverbanks.
Slope Instability & Blasting	Heavy excavation for multi-lane highways triggers landslips and weakens	Mandate eco-sensitive tunnel designs and ban heavy dynamite blasting in

Environmental Stressor	Impact on Himalayan Ecosystem	Recommended Policy Rethink
	steep mountain bedrock.	fragile zones.
Solid Waste & Plastic Dumping	Millions of plastic bottles block mountain streams and clog natural drainage pathways.	Implement strict zero-plastic mandates and mandatory waste takeaway rules for tourists.
Over-Extraction of Groundwater	Surging hotel demand drains local springs and aquifers, causing land subsidence (e.g., Joshimath).	Set scientifically determined tourist carrying-capacity caps per district.

Transboundary Alert Systems & International Cooperation

The Himalayas span multiple sovereign nations—Nepal, India, China, Bhutan, and Pakistan. Because major river systems (such as the Kosi, Gandak, Indus, and Brahmaputra) cross international borders, hydroclimatic hazards do not stop at national boundaries.

Prof. Pokhrel underscored that effective disaster mitigation requires automated, real-time sensor networks placed at high altitudes, coupled with transparent data sharing between neighboring countries. Satellite radar monitoring (such as Sentinel and NISAR) must be integrated into municipal early warning systems to give downstream populations critical lead time to evacuate.

Conclusion: Moving from Crisis Response to Proactive Resilience

As climate change shifts monsoon intensity and accelerates glacial melting, continuing business-as-usual tourism and unregulated urban expansion in the Himalayas is a recipe for escalating disaster. Re-evaluating mountain tourism is no longer optional—it is a vital imperative to safeguard lives, infrastructure, and the ecological backbone of South Asia.